

Mechanical Evaluation of 3D-Printed Hand Prosthesis Joints

1st Denisse Uscanga-Tlalpan

Dept. of Electronics, Mechatronics and Computing
Universidad de las Américas Puebla
Cholula, Puebla, México
denisse.uscanga@udlap.mx

2nd Gerardo Diaz-Arango

Facultad de Instrumentación Electrónica
Universidad Veracruzana
Xalapa, Veracruz, México
guda.diaz.gd@gmail.com

Abstract—Upper limb amputations are regarded a major challenge for rehabilitation, affecting independence and quality of life. Prostheses are artificial devices designed to replace missing body parts, and their effectiveness is based on functional adaptation to patient needs. In this work, STL models of hand prostheses incorporating two articulation systems were evaluated: direct joints and flexible TPU bands. The fingers were manufactured from PLA, the forearm structure was produced in PET-G, and finger flexion was achieved by the use of flexible and rigid tensioners. Mechanical testing involved repeated 10-second loading and deformation was analyzed using GOM Correlate software. The mechanical performance of both systems was compared to assess structural integrity and functional efficiency.

Index Terms—hand prostheses, 3D printing, mechanical deformation

I. INTRODUCTION

Upper limb amputations are considered a major challenge for rehabilitation and daily functioning. According to a study published in *Prosthetics and Orthotics International* (2021), it was estimated that by 2017, approximately 11.3 million individuals around the world lived with unilateral upper limb amputations. In addition, around 11.0 million people were reported to have experienced bilateral upper limb amputations, which highlights not only the prevalence of such cases but also their profound impact on independence and quality of life. This distribution is regarded as evidence of both the diversity and complexity of amputations worldwide, as well as the need for personalized approaches to rehabilitation and functional support for each patient [1]. A prosthesis is defined as an artificial medical device designed to replace a missing body part. To fulfill its intended purpose, it must be appropriately adapted and functionally effective in meeting the comprehensive needs of the user [2]. The remainder of this paper is structured as follows: Section II provides a summary of the state of the art regarding prosthesis types. Section III presents the details of the implementation of two 3D hand prosthesis models. In Section IV, the results are analyzed and discussed. Finally, Section V outlines the main conclusions of the study.

Identify applicable funding agency here. If none, delete this.

II. STATE OF THE ART

Prosthetic design is commonly characterized by the classification of devices into passive and active types, the diversity of activation mechanisms, and the selection of appropriate materials.

A. Passive Prostheses

Passive prosthetic devices (Figure 1) prioritize appearance and comfort, offering low cost cosmetic benefits despite limited functionality. They are widely used, with about one in three amputees adopting them, often as an initial option after amputation [2], [3].



Fig. 1. Components of a prosthetic arm device [2].

B. Active Prostheses

An active prosthesis generates movement using external power, such as electric motors or mechanical systems, allowing controlled functions of motion through myoelectric or body-controlled mechanisms [4].

1) *Body-Powered Devices (BPD)*: A body-powered prosthesis is controlled by the user's body movements, typically through a harness and cable system that transmits bodily force to the prosthesis to activate its function. An example is a body-powered upper limb prosthesis, which is controlled by a harness connected via a cable to the healthy shoulder. This system includes a socket, a wrist, a control cable, and a terminal device (such as a hand or hook) [2], as shown in Figure 2.

2) *Myoelectric Prosthesis*: Upper limb myoelectric prostheses (Figure 3) operate through the use of electric motors powered by an external energy source. Joint movement is controlled through the muscle activity of the residual limb.

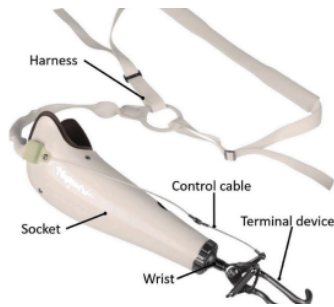


Fig. 2. Body-powered upper limb prosthesis [4].

Electromyographic (EMG) signals from the residual limb are detected by surface electrodes, amplified, and then processed by a controller that activates battery-powered motors to move the hand, wrist, or elbow [2].

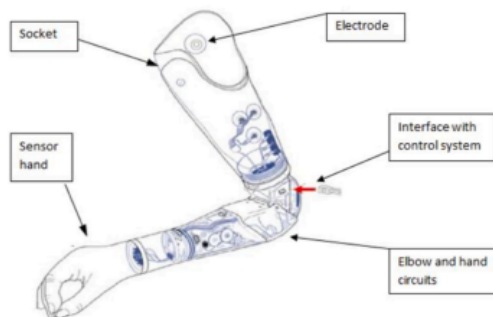


Fig. 3. Diagram of a myoelectric arm prosthesis [7].

3) *Hybrid Prostheses*: A hybrid upper extremity prosthesis combines myoelectric and body-powered components to leverage the benefits of both systems. Despite their advantages, harness use can be uncomfortable and demanding, particularly at short transhumeral levels [2], [6].

C. Anatomical Structure of the Hand

The human hand is a complex and highly specialized structure. It is composed of 27 bones, 27 joints, 34 muscles, more than 100 ligaments and tendons, as well as an extensive network of blood vessels and nerves [8].

1) *General Division of the Hand*: The hand can be divided into three main regions [8]:

- **Wrist (carpus)**: Contains the carpal bones and forms the joint with the forearm.
- **Palm (metacarpus)**: Connects the wrist bones to the fingers.
- **Fingers (phalanges)**: Allow fine movements.

2) *Bones and Joints*: As shown in Figure 4, the wrist consists of eight carpal bones that articulate with the forearm, while five metacarpals connect to the fingers through the **MCP joints**. Each finger has three phalanges separated by **PIP** and **DIP** joints, except for the thumb, which has two phalanges and a single interphalangeal joint [8].

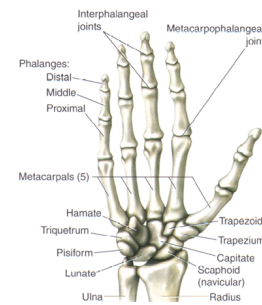


Fig. 4. Anterior view of the hand with the carpal bones, metacarpals, and phalanges, along with their main joints [10].

3) *Soft Tissues: Articular cartilage*: Covers the ends of bones at each joint, allowing smooth movement and cushioning impacts [9].

Ligaments: As shown in Figure 5, these tough, rope-like ligaments connect bones and stabilize finger joints. Each joint has two collateral ligaments, with the volar ligament, the strongest in the hand, limiting hyperextension of the PIP joint [9].

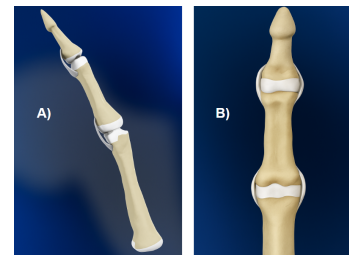


Fig. 5. Ligaments: A) volar and B) collateral of the phalangeal joints [10].

4) *Hand Muscles*: Muscles allow movement of the hand through their contraction and can be classified as [9]:

- **Intrinsic Muscles**: Originate within the hand and control fine movements.
- **Extrinsic Muscles**: Originate in the forearm and elbow, enabling wide hand and wrist movements. Each finger is controlled by three intrinsic and three extrinsic muscles, with the index and pinky having an additional extrinsic extensor.

5) *Hand Tendons*: Tendons connect muscles to bones. When the muscle contracts, the tendon pulls on the bone and produces movement. As illustrated in Figure 6, there are two main types [9]:

- **Flexor tendons**: Located on the palmar side of the hand; they allow the fingers to bend.
- **Extensor tendons**: Located on the dorsal side of the hand; they allow the fingers to extend.

D. Technologies for the Design and Development of Hand Prostheses

1) *3D Printing*: 3D printing, also known as additive manufacturing, is a manufacturing process that allows the creation

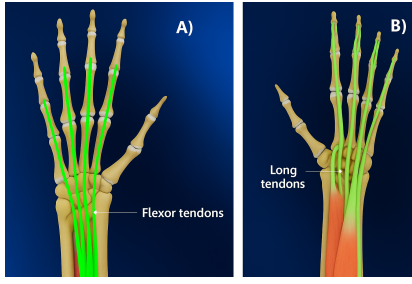


Fig. 6. Tendons of the hand: A) flexors and B) extensors [10].

of three-dimensional objects through the successive deposition of layers of material, guided by a digital model. This technology uses computer-aided designs (CAD) to build parts with complex geometries that would be difficult or impossible to achieve using traditional methods [11].

2) *Transmission Elements*: The transmission elements in hand prostheses are the mechanisms responsible for transferring the movement generated by the actuators to the joints and finger segments, as shown in Table I.

TABLE I
SUMMARY OF TRANSMISSION ELEMENTS USED IN HAND PROSTHESES.

Element	Description
Tendon-like cables	Imitate human tendons, transmitting force over distances; lightweight but with some friction.
Gears	Provide precise torque transmission; may exhibit mechanical backlash.
Pulleys and belts	Low-maintenance and suitable for compact designs.
Cranks and connecting rods	Convert rotational motion to linear motion; useful for articulated mechanisms.
Elastic or passive joints	Absorb impacts, allow passive adaptation, and enhance human–robot interaction.

E. Mechanical Deformation Metrics in Transmission Elements

1) *Unit Strain*: The **unit strain** (ε) measures the relative change in the length of a material when a load is applied. It is a dimensionless quantity that indicates how much an object is stretched or compressed compared to its original length [17].

$$\varepsilon = \frac{\Delta L}{L_0} \quad (1)$$

Where:

- ΔL is the change in the material's length (mm, m).
- L_0 is the original length of the material before loading (mm, m).

If the strain is reported as a percentage, divide by 100 to obtain ε [17]:

$$\varepsilon = \frac{\text{Strain (\%)}}{100} \quad (2)$$

2) *Young's Modulus*: Young's modulus, also called the modulus of elasticity, is a fundamental mechanical property that describes the stiffness of a material. It is defined as the ratio of stress (applied force per unit area) to unit strain

(relative change in length) in the elastic region of the material [18].

3) *Hooke's Stress*: The **Hooke's stress** (σ) describes the internal resistance of a material when subjected to an external load. It is based on **Hooke's law**, which establishes a linear relationship between stress and strain in elastic materials.

$$\sigma = E \cdot \varepsilon \quad (3)$$

Where:

- σ is the stress (Pa, MPa, GPa).
- E is the Young's modulus of the material (Pa, MPa, GPa), which represents its stiffness.
- ε is the unit strain (dimensionless).

Through this calculation, it is possible to determine how a material responds to applied loading within the elastic range, before permanent deformation occurs [19].

III. DESIGN AND IMPLEMENTATION

A. Materials

Through the design and modification of CAD models of hand prostheses, a comparison was made between two different types of joints present in the corresponding models. One of the prostheses is of the body-powered type, characterized by fixed joints arranged in chains, while the other is of the robotic type, in which the joints have flexible materials between the connecting elements. The corresponding materials are shown in Table II.

TABLE II
MATERIALS USED IN HAND PROSTHESES ACCORDING TO THE TYPE OF PROSTHESES.

Prosthesis type	Materials
Body-powered	Fingers: white PLA; Palm and forearm: PET-G; Tension cables: SPECTRA Extreme Braid — 30 lb fishing line; Flexible cables: elastic silicone-type cable coated with polyester.
Robotic	Fingers and palm: white PLA; Forearm and servomotor housing: PET-G; Joints: TPU; Tension cables: SPECTRA Extreme Braid — 30 lb fishing line; Electronic system: FSR402 pressure sensor, TURNIGY 11.1 V 1000 mAh battery, LM2596 voltage regulator, Arduino Nano, MG995 servomotor, 10 k Ω resistor.

Figure 7 shows the schematic of the electronic circuit used in the robotic prosthesis model. The circuit is activated by an FSR402 pressure sensor, triggering the servomotor and the subsequent movement of the fingers.

B. Printing Specifications

The parts corresponding to each prosthesis model were printed using the Flashforge Adventurer 5M Pro 3D printer with Orca Slicer. The printing configuration specifications are presented in Tables III and IV. For the body-powered hand, the index finger was printed in different positions, while the flexible joints of the robotic prosthesis were printed with different infill percentages.

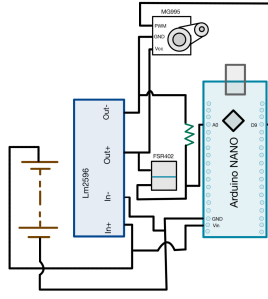


Fig. 7. Schematic of the electronic circuit used in the robotic prosthesis model.

TABLE III
SPECIFICATIONS OF 3D PRINTING FOR THE BODY-POWERED HAND COMPONENTS.

Component	Infill (%)	Layer (mm)	Height	Printing Orientation / Notes
Forearm	60	0.12		Vertical, tree supports
Fingers	80	0.12		Vertical, horizontal (up/down), and 45° incline; tree supports
Palm	60	0.12		Vertical, tree supports; supports modified to avoid obstruction of pinky tendon holes
Servo box	20	0.12		Horizontal, tree supports; tabs removed and holes placed on walls for mounting on forearm

C. Mechanical Analysis

For the mechanical analysis, GOM Correlate software was used to record the finger parts of each prosthesis model (robotic and body-powered) under constant compression for 10 seconds. Marked points enabled the software to track deformations and calculate unit strain and Hooke's stress using (2) and (3). The stress–time results were processed, and the time axis was adjusted to align datasets, generating comparative plots in MATLAB. The analysis was performed from external lateral views, as shown in Figure 8.

TABLE IV
SPECIFICATIONS OF 3D PRINTING FOR THE ROBOTIC HAND COMPONENTS.

Component	Infill (%)	Layer (mm)	Height	Printing Orientation / Notes
Forearm	60	0.12		Vertical, no supports; aesthetic finish
Palm	60	0.12		Vertical, tree supports; thumb area free
Fingers	80	0.12		Vertical, tree supports; space for flexible joints
Joints	20–50	0.12		Horizontal, raft base; base prevents layer detachment and ensures flexibility
Servo boxes	20	0.12		Horizontal, tree supports; preserves holes and tabs; printed with lids

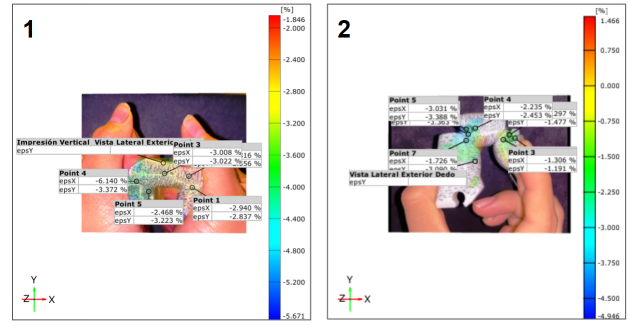


Fig. 8. Lateral view perspectives in the GOM Correlate software: (1) body-powered prosthesis, (2) robotic prosthesis.

IV. ANALYSIS OF RESULTS

A. Printing Results

In Figure 9, the printed and assembled components of both prosthesis models (body-powered and robotic) are presented. On the left, the structural elements of the body-powered model can be observed, while the central section shows the robotic hand prototype with integrated flexible joints. On the right, the electronic system used for actuation is displayed, consisting of a pressure sensor (FSR402), a microcontroller (Arduino Nano), and a servomotor, which together enable the mechanical response of the robotic hand.

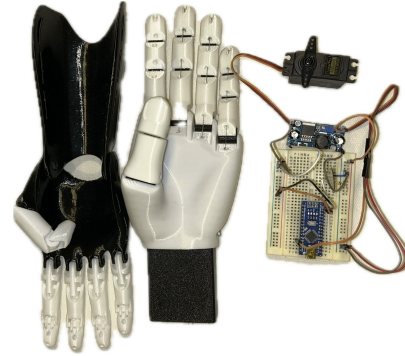


Fig. 9. 3D-printed components of the body-powered and robotic prosthesis models, along with the electronic circuit.

B. Mechanical Results

Body-Powered Prosthesis: Through strain analysis in GOM, Figures 10 and 11 show Hooke's stresses at inspection point 2 of the PLA material, located between the distal and middle phalanges. The results demonstrate that the printing orientation directly influences the mechanical response of the component: vertical printing exhibited the most negative stress values, reflecting lower resistance in this area; downward horizontal printing showed intermediate performance, with negative stresses but less pronounced; the 45° inclined printing displayed an irregular distribution with significant stress variations; whereas upward horizontal printing demonstrated the best performance, with values closer to zero and even positive,

suggesting greater resistance and structural stability at this critical point.

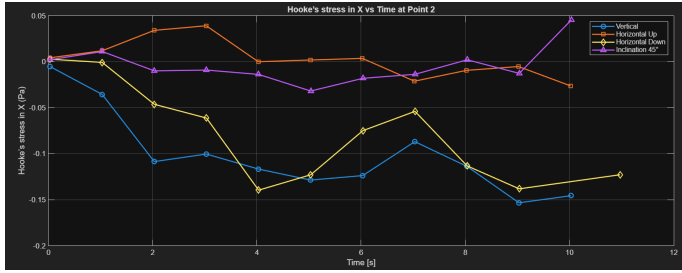


Fig. 10. Hooke's stress in X observed between the distal and middle phalanges under repeated loading for 10 seconds in PLA material for the body-powered prosthesis.

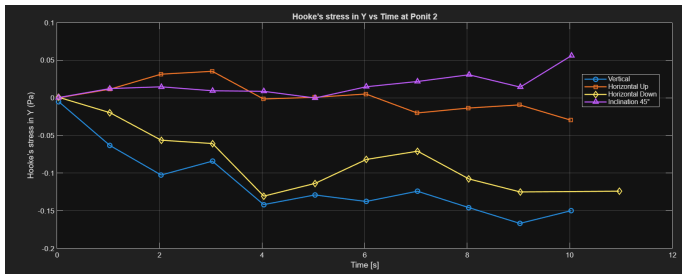


Fig. 11. Hooke's stress in Y observed between the distal and middle phalanges under repeated loading for 10 seconds in PLA material for the body-powered prosthesis.

On the other hand, Figures 12 and 13 show Hooke's stresses at inspection point 4, located between the middle and proximal phalanges. The vertical printing condition generates the highest compressive stresses, reaching values close to -0.3 Pa in X and -0.22 Pa in Y , indicating greater load transfer in this alignment. In contrast, the horizontal-up printing orientation induces almost no stresses, remaining near zero with slight tensile variations, while the horizontal-down orientation produces moderate compressions (between -0.05 and -0.15 Pa) with oscillations in both axes. Finally, the 45° inclined orientation maintains low stresses close to zero, with slight tensile tendencies toward the end, suggesting a more balanced distribution compared to the other orientations.

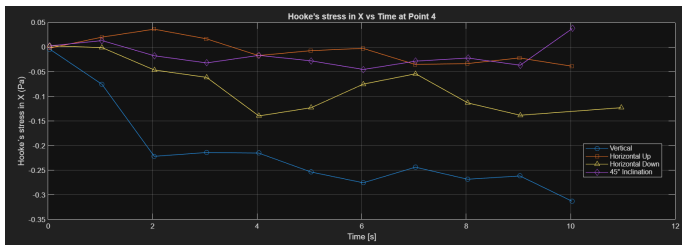


Fig. 12. Hooke's stress in X observed between the middle and proximal phalanges under repeated loading for 10 seconds in PLA material for the body-powered prosthesis.

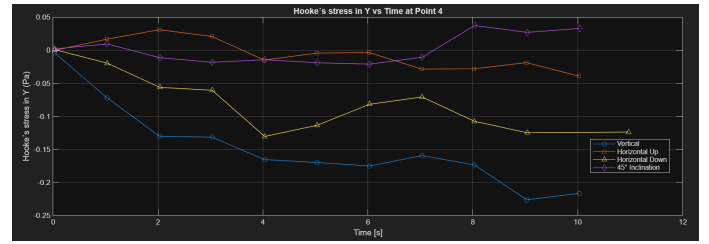


Fig. 13. Hooke's stress in Y observed between the middle and proximal phalanges under repeated loading for 10 seconds in PLA material for the body-powered prosthesis.

Robotic Prosthesis: For the robotic hand, at inspection point 2, located between the distal and middle phalanges with TPU material, it is observed that the infill percentage has a notable effect on Hooke's stresses. In Figure 14, corresponding to the X -axis, the 20% infill produces the highest tensile stress values, reaching approximately 3.2 Pa, while the 30% infill shows a tendency toward compressive stresses (up to -1.0 Pa). Meanwhile, the 40% and 50% infills maintain stresses close to zero, with moderate oscillations suggesting a more stable distribution. In Figure 15, corresponding to the Y -axis, the influence of infill is repeated: the 50% infill reaches the highest tensile stress values (around 2.8 Pa), followed by the 20%, while the 30% infill behaves oppositely, showing considerable compression (up to -3.5 Pa). Overall, these results indicate that higher infill levels tend to favor tensile resistance, while intermediate values such as 30% may induce significant compressive behavior at this point.

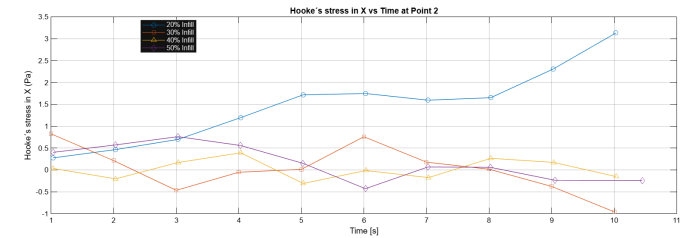


Fig. 14. Hooke's stress in X observed between the middle and distal phalanges under repeated loading for 10 seconds in TPU material for the robotic prosthesis.

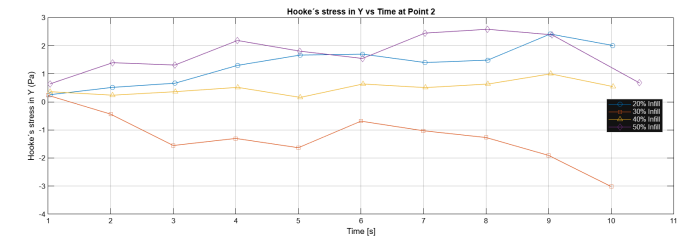


Fig. 15. Hooke's stress in Y observed between the middle and distal phalanges under repeated loading for 10 seconds in TPU material for the robotic prosthesis.

At inspection point 5, located between the middle and proximal phalanges, Figure 16 shows that along the X -axis the

20% infill reaches the highest tensile values (≈ 2.1 Pa), while the 30% infill exhibits high compressive stresses (≈ -4.0 Pa), indicating critical behavior under this condition. The 40% and 50% infill remain close to zero with slight oscillations, reflecting greater stability. Similarly, Figure 17, corresponding to the Y-axis, shows that the 20% infill generates the highest tensile value (≈ 2.8 Pa), the 30% infill induces significant compressive stresses (≈ -4.5 Pa), and the higher percentages (40% and 50%) exhibit a more balanced behavior, with moderate and stable values. Overall, the results indicate that low infill (20%) enhances tensile response, whereas 30% infill leads to compressive instability, and higher infill levels (40–50%) promote better stress distribution.

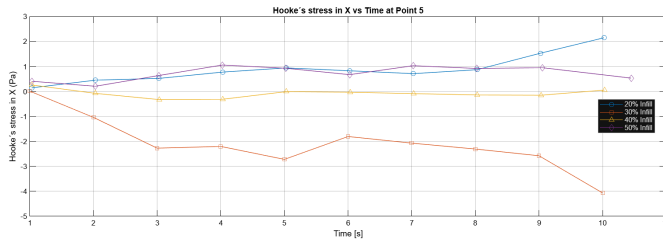


Fig. 16. Hooke's stress in X observed between the middle and proximal phalanges under repeated loading for 10 seconds in TPU material for the robotic prosthesis.

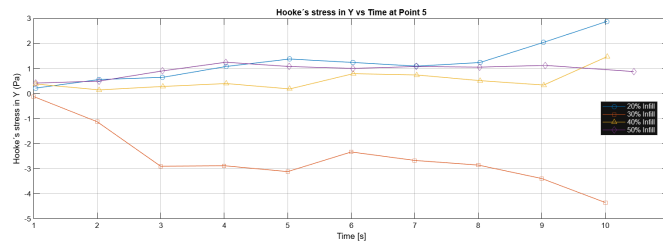


Fig. 17. Hooke's stress in Y observed between the middle and proximal phalanges under repeated loading for 10 seconds in TPU material for the robotic prosthesis.

V. CONCLUDING REMARKS

Mechanical analysis of prosthetic hand models demonstrates that both material selection and design parameters, such as printing orientation and infill percentage, significantly influence structural performance and, consequently, functional behavior. For the body-powered prosthesis, vertical printing led to higher compressive stresses and lower structural stability, while upward horizontal and 45° inclined orientations exhibited more balanced stress distributions, suggesting improved mechanical reliability under repetitive loading. In the robotic prosthesis, the infill percentage played a critical role: low infill levels (20%) improved the tensile response, intermediate values (30%) induced compressive instability, and higher infill percentages (40–50%) promoted a more uniform and stable stress distribution.

The study presents certain limitations that should be addressed in future work, such as the number of samples

analyzed for each model. Additionally, variations in filament brands, differences in material drying procedures, and possible moisture absorption during printing could have influenced the mechanical response of the specimens. The long-term mechanical wear of the printed components was not evaluated; therefore, durability over extended operational cycles remains uncertain. Future research should include a larger number of samples, controlled material pre-treatment processes, and long-term fatigue testing to better assess the mechanical degradation and functional performance of prosthetic components under realistic usage conditions.

REFERENCES

- [1] C. L. McDonald, S. Westcott-McCoy, M. R. Weaver, J. Haagsma, and D. Kartin, "Global prevalence of traumatic non-fatal limb amputation," *Prosthetics and Orthotics International*, vol. 45, 2021, doi: 10.1177/0309364620972258. [Online]. Available: https://journals.lww.com/poijournal/abstract/2021/04000/global_prevalence_of_traumatic_non_fatal_limb4.aspx
- [2] R. Brack and E. H. Amalu, "A review of technology, materials and R&D challenges of upper limb prosthesis for improved user suitability," *Journal of Orthopaedics*, vol. 23, 2021, doi: 10.1016/j.jor.2021.01.015. [Online]. Available: <https://doi.org/10.1016/j.jor.2021.01.015>
- [3] B. Maat, G. Smit, D. Plettenburg, and P. Breedveld, "Passive prosthetic hands and tools: a literature review," *Prosthetics and Orthotics International*, vol. 42, 2017, doi: 10.1177/0309364617710417.
- [4] S. Hussain, S. Shams, and S. Jawaid Khan, "Impact of medical advancement: Prostheses," *Computer Architecture in Industrial, Biomechanical and Biomedical Engineering*, 2019.
- [5] J. Webster, *Medical Instrumentation: Application and Design*, Wiley, 3rd ed., 2020.
- [6] E. Guzmán and G. Méndez, "Electromiografía en las ciencias de la rehabilitación," *Salud, Barranquilla*, vol. 34, 2018. [Online]. Available: http://www.scielo.org.co/scielo.php?script=sci_arttext&pid=S0120-55522018000300753
- [7] J. Uellendahl, "Myoelectric versus body-powered upper-limb prostheses," *Journal of Prosthetics and Orthotics*, vol. 29, pp. P25–P29, 2017.
- [8] M. Paul, "Sensory feedback system for prosthetic arms," University of Salford, 2015.
- [9] M. Latarjet and A. Ruiz Liard, *Anatomía Humana*. Buenos Aires: Editorial Médica Panamericana, 2020.
- [10] Grupo Ortopédico Edison-Metuchen, "Normal Hand Anatomy," 2020. [Online]. Available: <https://www.emog.net/normal-hand-anatomy/>
- [11] E. Gómez and C. Fernández, *Impresión 3D en cirugía ortopédica*. Madrid: Editorial Médica Panamericana, 2018.
- [12] Q. Zhang, D. Sun, W. Qian, and X. Xiao, "Modeling and control of a cable-driven rotary series elastic actuator for an upper limb rehabilitation robot," *Front. Neurobot.*, vol. 14, p. 13, Feb. 2020, doi: 10.3389/fnbot.2020.00013.
- [13] C. Della Santina, C. Piazza, G. Grioli, and A. Bicchi, "A review on the control of soft robotic hands," *Front. Neurobot.*, vol. 15, p. 664062, 2021.
- [14] G. Hirzinger *et al.*, "DLR's torque-controlled light weight robot III," in *Proceedings of the 2001 IEEE International Conference on Robotics and Automation (ICRA)*, 2001.
- [15] J. L. Pons, *Wearable Robots: Biomechatronic Exoskeletons*. Hoboken, NJ: John Wiley & Sons, 2008.
- [16] S. B. Godfrey *et al.*, "Review of compliant transmission elements in assistive robotic hands," *Technologies*, vol. 13, no. 1, p. 37, 2023.
- [17] R. C. Hibbeler, *Mecánica de Materiales*. México: Pearson Educación, 2006.
- [18] A. M. Colliou *et al.*, *Propiedades mecánicas y térmicas de los materiales*. Barcelona: Reverté, 1977.
- [19] J. Jiménez *et al.*, *Fundamentos de Elasticidad y Resistencia de Materiales*. Madrid: Ediciones Paraninfo, 2020.